CTU training solutions | Private Tertiary Education

CAPSTONE PROJECT

FINAL SYSTEM DEMONSTRATION SCRIPT

Student / Group Template

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
Group Members	Freek Schaefer-Gerber, Wian Sandenbergh
Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
Week / Stage	Final System Demonstration
Submission Date	29 October 2026

CTU Training Solutions (PTY) Ltd is fully accredited by the MICT SETA (ACC/2006/07/186) through the Quality Council for Trades and Occupations (QCTO). CTU Training Solutions is registered with the Department of Higher Education and Training as a Private College (2018/FE07/004) and provisionally registered as a private higher education institution (No 2014/HE07/004) Co. Reg No:2005/038944/07

1. Opening

Introduce the problem, project title and team.

Freek : "Good morning, Mr. Dunstan. We are Freek and Wian, and today we are presenting SmartDesk, a customer support ticketing system designed specifically for small IT support teams."

2. Problem and Objectives

Explain the problem and project objectives briefly.

Wian : "Small IT support teams often rely on shared inboxes, WhatsApp, or spreadsheets to manage requests. This leads to lost tickets, unclear ownership, and a lack of visibility into backlog. Our objective was to build a secure, centralised web application that enforces customer data isolation, allows staff to assign and track tickets, and provides a clear dashboard for workload visibility."

3. Architecture / Design

Show the main design and technologies.

Freek : "The system is built on a three-layer architecture.

- Frontend: Server-rendered Jinja2 HTML templates and CSS.
- Backend: Python with the Flask framework, handling routing, server-side validation, and session management.
- Database: A relational SQLite database.
- Security: We implemented Werkzeug password hashing, CSRF tokens on all POST forms, parameterised SQL queries to prevent injection, and strict server-side ownership checks to prevent cross-customer access."

4. Live Demonstration Flow

List the exact features to demonstrate in sequence, including expected results.

Wian : Step 1: Customer Workflow & Validation

- Action: Log in as the Customer (`customer@smartdesk.local`).
- Show: The Customer Dashboard (showing only their own tickets).
- Action: Click "Create Ticket". Try to submit it with an empty title.

CTU Training Solutions (PTY) Ltd is fully accredited by the MICT SETA (ACC/2006/07/186) through the Quality Council for Trades and Occupations (QCTO). CTU Training Solutions is registered with the Department of Higher Education and Training as a Private College (2018/FE07/004) and provisionally registered as a private higher education institution (No 2014/HE07/004) Co. Reg No:2005/038944/07

- Expected Result: The form rejects the submission and displays a clear validation message naming the field and the fix.
- Action: Fill in a valid title, description, category, and priority. Submit.
- Expected Result: A success message appears with the new Ticket ID.

Step 2: Staff Workflow & Assignment

- Action: Log out and log in as the Staff/Agent (`agent@smartdesk.local`).
- Show: The Agent Dashboard.
- Action: Open the unassigned ticket just created. Assign it to themselves.
- Action: Change the status from "Open" to "In Progress".
- Action: Post a comment replying to the customer.
- Expected Result: The assignee updates, status changes, and the comment appears chronologically with the agent's name and timestamp.

Step 3: Security / Data Isolation Test (Crucial)

- Action: Log out and log back in as a *different* Customer.
- Action: Paste the URL of the first customer's ticket into the browser bar.
- Expected Result: The system blocks access and returns a 403 Forbidden page, proving the `get_ticket()` ownership check works.

Step 4: Administrator Dashboard

- Action: Log in as the Admin (`admin@smartdesk.local`).
- Show: The Admin Dashboard.
- Expected Result: Point out the status counts (Open, In Progress, Resolved). Note how priority is the only colour-coded element (High is red) to make triage fast.

5. Testing Evidence

Show selected test results and how defects were addressed.

Freek : "To ensure the system was robust, we ran 3 automated `pytest` functions and 20 manual test cases. During testing, we found and fixed 5 defects. Most notably, we discovered a critical vulnerability

where a customer could change a URL ID to view another customer's ticket. We resolved this by enforcing an ownership check on every route, which you just saw in Step 3. We also added CSRF tokens to fix a high-severity security issue."

6. Conclusion

State what was achieved, limitations and future improvements.

Wian : "SmartDesk successfully meets its MVP requirements. It provides a secure, validated, and isolated ticketing workflow. While features like email notifications, file attachments, and a formal audit history were out of scope for this MVP, the system provides a strong, secure foundation. Future work would include adding those features, as well as pagination for the dashboard to handle thousands of tickets."

7. Backup Plan

Record screenshots/video/test data to use if a technical issue prevents live demonstration.

- Technical Failure: If the local server crashes or Wi-Fi drops, we have a full screen-recorded video of the exact demonstration flow (Steps 1-4) ready to play.
- Database Failure: We have the `flask init-db` and `flask seed-demo` commands ready in a terminal tab to instantly rebuild the database, along with a backup copy of the `support.db` file.
- Visuals: A PDF containing high-resolution screenshots of every screen (Dashboard, Create Ticket, Validation Errors, 403 Forbidden, Comments) is open on a secondary screen.